

16. Surface area and Volume

Standard unit of volume :-

1) Volume of cuboid = $l \times b \times h$

2) Volume of cube = Side^3

-: Practice Set 16.1 :-

1. Find the volume of a box if its length, breadth and height are 20 cm, 10.5 cm and 8 cm respectively.

→ $l = 20 \text{ cm}, b = 10.5 \text{ cm}, h = 8 \text{ cm}$

Volume of box = $l \times b \times h$

$= 20 \times 10.5 \times 8$

$= 210 \times 8$

$= 1680 \text{ cm}^3$

2. A cuboid shape soap box has volume 150 cc. Find its thickness if its length is 10 cm and breadth is 5 cm.

→ Volume of cuboid = $l \times b \times h$

$150 = 10 \times 5 \times h$

$150 = 50 \times h$

$h = \frac{150}{50}$

$\therefore h = 3 \text{ cm}$

4. For rain water harvesting a tank of length 10 m, breadth 6 m and depth 3 m is built. What is the capacity of the tank? How many litres of water can it hold?

→ $l = 10 \text{ m}, b = 6 \text{ m}, d = 3 \text{ m}$

$\therefore h = 3 \text{ m}$

$$\begin{aligned}\text{Volume of tank} &= l \times b \times h \\ &= 10 \times 6 \times 3 \\ &= 180 \text{ m}^3\end{aligned}$$

$\therefore$ Capacity of tank is 180 m^3
The tank will hold 1,80,000 litre water

Surface area of cylinder

1] Curved surface area of cylinder $= 2\pi r h$

2] Total surface area $= 2\pi r (h + r)$

—: Practice Set 16.2 :—

1] In each example given below, radius of base of a cylinder and its height are given. Then find the curved surface area and total surface area.

ii] $r = 7 \text{ cm}$, $h = 10 \text{ cm}$

$$\begin{aligned}\rightarrow \text{Curved surface Area} &= 2\pi r h \\ &= 2 \times 22 \times 7 \times 10 \\ &= 44 \times 10\end{aligned}$$

Curved surface Area $= 440 \text{ cm}^2$

$$\begin{aligned}\text{Total surface Area} &= 2\pi r (h + r) \\ &= 2 \times 22 \times 7 (10 + 7) \\ &= 44 (17) \\ &= 748 \text{ sq. cm}\end{aligned}$$

2) $r = 1.4 \text{ cm}$, $h = 2.1 \text{ cm}$

→ Curved surface Area = $2\pi rh$

$$= \frac{2 \times 22 \times 1.4 \times 2.1}{7}$$

$$= 2 \times 22 \times 0.2 \times 2.1$$

$$= 2 \times 22 \times 0.42$$

$$= 22 \times 0.84$$

Curved surface Area = 18.48 sq cm

Total surface Area = $2\pi r(h+r)$

$$= \frac{2 \times 22 \times 1.4 (2.1 + 1.4)}{7}$$

$$= \frac{2 \times 22 \times 1.4 (3.5)}{7}$$

$$= 2 \times 22 \times 0.2 (3.5)$$

$$= 2 \times 22 \times 0.7 (3.5)$$

$$= 8.8 (3.5)$$

Total surface Area = 30.80 cm^2

3) $r = 2.5 \text{ cm}$, $h = 7 \text{ cm}$

→ Curved surface Area = $2\pi rh$

$$= \frac{2 \times 22 \times 2.5 \times 7}{7}$$

$$= 2 \times 22 \times 2.5$$

$$= 44 \times 2.5$$

Curved surface Area = 110.0 cm^2

Total surface Area = $2\pi r(h+r)$

$$= \frac{2 \times 22 \times 2.5 (7 + 2.5)}{7}$$

$$= \frac{2 \times 22 \times 2.5 \times (9.5)}{7}$$

$$\begin{aligned}
 &= \frac{44 \times 2.5 \times (9.5)}{7} \\
 &= \frac{110 \times (9.5)}{7} \\
 &= \frac{10450}{7}
 \end{aligned}$$

$$\text{Total Surface Area} = 149.29$$

$$\text{Q1 } r = 70 \text{ cm}, h = 1.4 \text{ cm}$$

$$\begin{aligned}
 \rightarrow \text{Curved surface Area} &= 2\pi r h \\
 &= 2 \times 22 \times 70 \times 1.4 \\
 &= 2 \times 22 \times 10 \times 1.4 \\
 &= 44 \times 10 \times 1.4 \\
 &= 440 \times 1.4
 \end{aligned}$$

$$\text{Curved surface Area} = 616.0 \text{ cm}^2$$

$$\begin{aligned}
 \text{Total surface Area} &= 2\pi r (h + r) \\
 &= 2 \times 22 \times 70 (1.4 + 70) \\
 &= 2 \times 22 \times 10 (71.4) \\
 &= 440 \times (71.4)
 \end{aligned}$$

$$\text{Total surface Area} = 31416.0 \text{ cm}^2$$

$$\text{Q2 } r = 4.2 \text{ cm}, h = 14 \text{ cm}$$

$$\begin{aligned}
 \rightarrow \text{Curved surface Area} &= 2\pi r h \\
 &= 2 \times 22 \times 4.2 \times 14 \\
 &= 2 \times 22 \times 4.2 \times 2 \\
 &= 44 \times 4.2 \times 2
 \end{aligned}$$

$$\text{Curved Surface Area} = 369.6 \text{ cm}^2$$

$$\text{Total Surface Area} = 2\pi r(h+r)$$

$$= 2 \times 22 \times 4.2(14+4.2)$$

$$= 2 \times 22 \times 0.6(18.2)$$

$$= 44 \times 0.6(18.2)$$

$$= 26.4(18.2)$$

$$\text{Total Surface Area} = 480.48 \text{ cm}^2$$

Volume of cylinder :-

$$\text{Volume of cylinder} = \pi r^2 h$$

Practice Set 16.3

1. Find the volume of the cylinder if height (h) and radius of the base (r) are as given below.

(i) $r = 10.5 \text{ cm}, h = 8 \text{ cm}$

$$\rightarrow \text{Volume of Cylinder} = \pi r^2 h$$

$$= 22 \times 10.5 \times 10.5 \times 8$$

$$= 22 \times 10.5 \times 10.5 \times 8$$

$$= 33 \times 84$$

$$= 2772 \text{ cm}^3$$

(ii) $r = 2.5 \text{ m}, h = 7 \text{ m}$

→ Volume of cylinder = $\pi r^2 h$
 $= \frac{22}{7} \times 2.5 \times 2.5 \times 7$
 $= 22 \times 2.5 \times 2.5$
 $= 55 \times 2.5$
 $= 137.5 \text{ m}^3$

(3) $r = 4.2 \text{ cm}$, $h = 5 \text{ cm}$
 → Volume of cylinder = $\pi r^2 h$
 $= \frac{22}{7} \times 4.2 \times 4.2 \times 5$
 $= 22 \times 0.6 \times 4.2 \times 5$
 $= 13.2 \times 4.2 \times 5$
 $= 55.44 \times 5$
 $= 277.2 \text{ cm}^3$

(4) $r = 5.6 \text{ cm}$, $h = 5$
 → Volume of cylinder = $\pi r^2 h$
 $= \frac{22}{7} \times 5.6 \times 5.6 \times 5$
 $= 22 \times 0.8 \times 5.6 \times 5$
 $= 17.6 \times 5.6 \times 5$
 $= 98.56 \times 5$
 $= 492.8 \text{ cm}^3$

8x Hohl

2. Find the total surface area of a closed cylindrical drum if its diameter is 50 cm and height is 45 cm. (π = 3.14)
 → Radius = $\frac{\text{diameter}}{2} = \frac{50}{2} = 25$

Total sur

3. Find the surface
→ Curved

$$\begin{aligned}
 \text{Total surface Area} &= 2\pi r(h+r) \\
 &= 2 \times 22 \times 25(45+25) \\
 &= 2 \times 3.14 \times 25(70) \\
 &= 6.28 \times 157(70) \\
 &= 6.28 \times 25(70) \\
 &= 157 \times 70 \\
 &= 10990 \text{ cm}^2
 \end{aligned}$$

3. Find the area of base and radius of a cylinder if its curved surface area is 660 sq cm and height is 21 cm.

$$\begin{aligned}
 \rightarrow \text{Curved surface Area} &= 2\pi rh \\
 660 &= 2 \times 22 \times r \times 21
 \end{aligned}$$

$$660 = 2 \times 22 \times r \times 3$$

$$660 = 44 \times r \times 3$$

$$660 = 132 \times r$$

$$r = \frac{660}{132}$$

$$r = 5 \text{ cm}$$

$$\text{base} = \pi r^2$$

$$= 22 \times 5 \times 5$$

$$= 3.14 \times 5 \times 5$$

$$= 3.14 \times 5 \times 5$$

$$= 78.50 \text{ cm}^2$$

4. Find the area of the sheet required to make a cylindrical container which is open at one side and whose diameter is 28 cm and height is 20 cm. Find the

approximate area of the sheet required to make a lid of height 2 cm for this container.

→ Given: diameter = 28 cm, height = 20 cm
height of lid = 2 cm, radius = 14 cm

Curved surface Area of sheet required to make cylindrical container + base of cylindrical container

$$2\pi rh + \pi r^2$$

$$2 \times \frac{22}{7} \times 14 \times 20 + \frac{22}{7} \times 14 \times 14$$

$$= 2 \times 22 \times 2 \times 20 + 22 \times 2 \times 14$$

$$= 44 \times 2 \times 20 + 44 \times 14$$

$$= 88 \times 20 + 616$$

$$= 1760 + 616$$

$$= 2376 \text{ sq cm}$$

$$\text{Area of sheet} = \boxed{2376 \text{ sq cm}}$$

Area of sheet required to make lid = Curved surface Area of lid + base of height lid

$$= 2\pi rh + \pi r^2$$

$$= 2 \times \frac{22}{7} \times 14 \times 2 + \frac{22}{7} \times 14 \times 14$$

$$= 2 \times 22 \times 2 \times 2 + 22 \times 2 \times 14$$

$$= 44 \times 2 \times 2 + 44 \times 14$$

$$= 88 \times 2 + 616$$

$$= 176 + 616$$

$$\text{base of lid} = \boxed{792 \text{ sq cm}}$$

4. Find the volume of the cylinder if the circumference of the cylinder is 132 cm and height is 25 cm.

$$\text{Circumference} = 2\pi r$$

$$\rightarrow 132 = 2 \times \frac{22}{7} \times r$$

$$132 \times 7 = 44 \times r$$

$$924 = 44 \times r$$

$$r = \frac{924}{44}$$

$$r = 21 \text{ cm.}$$

Now,

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times 21 \times 21 \times 25$$

$$= 22 \times 3 \times 21 \times 25$$

$$= 66 \times 21 \times 25$$

$$= 66 \times 525$$

$$\text{Volume of cylinder} = \boxed{34650 \text{ cm}^3}$$

2.

Euler's Formula:

The formula regarding the faces (F), vertices (V), and edges (E) of solid figures is called Euler's formula.

The formula is represented as follows:-

$$F + V = E + 2$$

2. How much iron is needed to make rod of length 90 cm and diameter 1.4 cm?

$$\rightarrow \text{diameter} = 1.4, \text{ radius} = 0.7$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22 \times 0.7 \times 0.7 \times 90}{7}$$

$$= 22 \times 0.1 \times 0.7 \times 90$$

$$= 0.2 \times 0.7 \times 90$$

$$= 15.4 \times 90$$

$$\text{Volume of cylinder} = 1386 \text{ cm}^3$$

3. How much water will a tank hold if the interior diameter of the tank is 1.6 m and its depth is 0.7 m?

→ diameter = 1.6, radius = 0.8

depth = 0.7, ∴ height = 0.7

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22 \times 0.8 \times 0.8 \times 0.7}{7}$$

$$= 22 \times 0.8 \times 0.8 \times 0.1$$

$$= 0.76 \times 0.8 \times 0.1$$

$$= 0.1408 \times 0.1$$

$$= 0.1408$$

$$\text{Volume of cylinder} = 1.408$$

convert into litre = 1.408×1000

$$= 1408 \text{ L}$$